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Therapeutic, fitness, and sports enhancement device

Abstract

A therapeutic, fitness and sports enhancement device which includes a cylindrically shaped body having a predetermined density and predetermined diameter, the body including a plurality of projections of a predetermined shape. One or more of the predetermined density and the predetermined diameter in combination with the predetermined shape of the projections optimize mobilization of soft tissue structures of the human body.

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Field of Classification Search

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 18/902,214 filed on Sep. 30, 2024, which is a continuation application of U.S. patent application Ser. No. 16/365,043 filed on Mar. 26, 2019 (now U.S. Pat. No. 12,133,825), which is a continuation application of U.S. patent application Ser. No. 15/593,405 filed on May 12, 2017 (now U.S. Pat. No. 10,265,238), which claims benefit of and priority thereto under 35 U.S.C. § 119, 120, 363, 365 and 37 C.F.R. §§ 1.55 and 1.78 and which is incorporated herein by reference, and U.S. patent application Ser. No. 15/593,405 is a continuation application of U.S. patent application Ser. No. 15/005,380 filed on Jan. 25, 2016 (now U.S. Pat. No. 9,656,112), which claims benefit of and priority thereto under 35 U.S.C. § 119, 120, 363, 365 and 37 C.F.R. §§ 1.55 and 1.78 and which is incorporated herein by reference, and Ser. No. 15/005,380 is a divisional application of U.S. patent application Ser. No. 13/065,850 filed on Mar. 31, 2011 (now U.S. Pat. No. 9,345,921), which claims the benefit of and priority thereto under 35 U.S.C. § 119, 120, 363, 365 and 37 C.F.R. §§ 1.55 and 1.78 and which is incorporated herein by reference, and Ser. No. 13/065,850 is a continuation-in-part of U.S. patent application Ser. No. 11/827,361 filed on Jul. 11, 2007 (now U.S. Pat. No. 7,918,774), which claims the benefit of and priority to U.S. Provisional Application Ser. No. 60/831,726 filed Jul. 18, 2006, each of which is incorporated by reference herein.

FIELD OF THE INVENTION

(1) This subject invention relates to a therapeutic, fitness, and sports enhancement device.

BACKGROUND OF THE INVENTION

(2) Conventional cylindrical foam rolls are often used for tissue mobilization, improving body core strength and balance training. When used for tissue mobilization, the user typically lies or sits on the cylindrical foam roll and moves back and forth to mobilize target soft tissue. The benefits of soft tissue mobilization include, inter alia, decreased muscle soreness, relief from joint stress, decreased neuromuscular hypertonicity, increased joint range motion, extensibility of musculotendinous junctions, increased neuromuscular efficiency, and maintenance of normal functional muscle length. These benefits enhance recovery of injuries and also maximize sports performance and result in improved fitness.

(3) However, the completely cylindrical shape of a conventional foam roll minimizes the effect of tissue mobilization. Moreover, such a design often results in the user falling off the cylindrically shaped foam roll, which is aggravating and cumbersome.

(4) When used for body core strength and balance training, the user typically stands or on the cylindrical foam roll device to enhance balance, coordination and core strength. However, because the user can easily fall off the cylindrical shaped foam roll, the effectiveness of the body core strength and balance training session is often minimized.

(5) Smaller sized cylindrical foam rolls may be used to mobilize small soft tissue structures, such as the foot, plantar fascia, heel, Achilles tendon and associated calf tissue, elbow, wrist, knee, head, neck, and the like, to assist in the recovery of plantar fasciitis or similar type ailments.

(6) However, conventional smaller sized cylindrical foam rolls are typically ineffective at mobilizing the soft tissue of small soft tissue structures because they cannot effectively break up collagenous fibers therein.

(7) Conventional foot massaging devices and patient positioning devices are known which may include rolls with projections or cone shaped rolls. See, e.g., U.S. Pat. Nos. 2,039,495, 5,359,739, 5,411,470, and 6,764,456. However, the rolls and devices disclosed therein are limited to a one diameter, one density, and a one type of shape of the projections and thus fail to effectively mobilize soft tissue structures of the human body.

BRIEF SUMMARY OF THE INVENTION

(8) It is therefore an object of this invention to provide a therapeutic, fitness, and sports enhancing device.

(9) It is a further object of this invention to provide such a therapeutic, fitness, and sports enhancement device in which one or more of the density and diameter in combination with the shape of the projections may be chosen to effectively mobilize soft tissue structures of the human body.

(10) It is a further object of this invention to provide such a therapeutic, fitness, and sports enhancement device which maximizes mobilization of soft tissue.

(11) It is a further object of this invention to provide such a therapeutic, fitness, and sports enhancement device which enhances flexibility.

(12) It is a further object of this invention to provide such a therapeutic, fitness, and sports enhancement device which improves body core strength training.

(13) It is a further object of this invention to provide such a therapeutic, fitness, and sports enhancement device which optimizes soft tissue distensibility.

(14) It is a further object of this invention to provide such a therapeutic, fitness, and sports enhancement device which improves balance training.

(15) It is a further object of this invention to provide such a therapeutic, fitness, and sports enhancement device which enhances recovery from injury.

(16) It is a further object of this invention to provide such a therapeutic, fitness, and sports enhancement device which maximizes sports performance.

(17) It is a further object of this invention to provide such a therapeutic, fitness, and sports enhancement device which improves fitness.

(18) It is a further object of this invention to provide such a therapeutic, fitness, and sports enhancement device which improves mobilization of the soft tissue of small soft tissue structures.

(19) It is a further object of this invention to provide such a therapeutic, fitness, and sports enhancement device which effectively breaks up collagenous fibers of soft tissue structures.

(20) It is a further object of this invention to provide such a therapeutic, fitness, and sports enhancement device which effectively helps treat plantar fasciitis.

(21) This invention features a therapeutic, fitness, and sports enhancement device including a cylindrically shaped body having a predetermined density and predetermined diameter, the body including a plurality of projections of a predetermined shape. One or more of the predetermined density and the predetermined diameter in combination with the predetermined shape of the projections optimize mobilization of soft tissue structures of the human body.

(22) In one embodiment, one or more of the predetermined density and the predetermined diameter in combination with the predetermined shape of the projections may optimize body core strength and optimize balance training. One or more of the predetermined density and the predetermined diameter in combination with the predetermined shape may effectively break up collagenous fibers of the soft tissue structures. One or more of the predetermined density and the predetermined diameter in combination with the predetermined shape of the projections may enhance flexibility of the soft tissue structures. One or more of the predetermined density and the predetermined diameter in combination with the predetermined shape of the projections may enhance distensibility of the soft tissue structures. The predetermined shape of the projections may include a rounded shape, a curved shape, a triangle shape, a square, a notch shape, a cylindrical shape with rounded ends, and cylindrical shape with flat ends. The projections may longitudinally extend over a length of the body. The projections may extend perpendicular to a longitudinal axis of the body. The projections may radially extend from the body at the same angles. The projections may radially extend from the body at different angles. The projections may be configured in a wave pattern. The projections may be configured in a Z pattern. The projections may be configured in a helical pattern. The projections may be individually spaced in a predetermined pattern over the body. The

predetermined pattern may include an aligned pattern of a plurality of shaped projections. The predetermined pattern may include an offset pattern of a plurality of shaped projections. The cylindrically shaped body and the projections may be made of a pliable material. The pliable material may include a closed-cell foam material. The closed-cell foam material may include a chemically cross-linked polyethylene foam. The pliable material may include expandable polyethylene. The pliable material may include plastic. The number of the projections may be configured to optimize mobilization of the soft tissue structures and/or optimize body core and balance training and/or effectively break up collagenous fibers of the soft tissue structures and/or enhance flexibility of the soft tissue structures and/or enhance distensibility of the soft tissue structures. The device may include a handle extending through the body.

(23) This invention also features a therapeutic, fitness, and sports enhancement device including a cylindrically shaped core having a predetermined diameter. An overlay is about the core having a predetermined density, the overlay including a plurality of projections of a predetermined shape. One or more of the predetermined density and the predetermined diameter in combination with the predetermined shape of the projections optimize mobilization of soft tissue structures of the human body.

(24) In one embodiment, one or more of the predetermined density and the predetermined diameter in combination with the predetermined shape of the projections may effectively optimize body core strength and optimize balance training. One or more of the predetermined density and the predetermined diameter in combination with the predetermined shape of the projections may effectively break up collagenous fibers of the soft tissue structures. One or more of the predetermined density and the predetermined diameter in combination with the predetermined shape of the projections may enhance flexibility of the soft tissue structures. One or more of the predetermined density and the predetermined diameter in combination with the predetermined shape of the projections may enhance distensibility of the soft tissue structures. The predetermined shape of the projections may include a rounded shape, a curved shape, a triangle shape, a square, a notch shape, a cylindrical shape with rounded ends, and cylindrical shape with flat ends. The projections longitudinally may extend over a length of the body. The projections may extend perpendicular to a longitudinal axis of the body. The projections may radially extend from the body at the same angles. The projections may radially extend from the body at different angles. The projections may be configured in a wave pattern. The projections may be configured in a Z pattern. The projections may be configured in a helical pattern. The projections may be individually spaced in a predetermined pattern over the body. The predetermined pattern may include an aligned pattern of a plurality of shaped projections. The predetermined pattern may include an offset pattern of a plurality of shaped projections. The cylindrically shaped body and the projections may be made of a pliable material. The pliable material may include a closed-cell foam material. The closed-cell foam material may include a chemically cross-linked polyethylene foam. The pliable material may include expandable polyethylene. The pliable material may include resin. The number of the projections may be configured to optimize mobilization of the soft tissue structures and/or optimize body core and balance training and/or effectively break up collagenous fibers of the soft tissue structures and/or enhance flexibility of the soft tissue structures and/or enhance distensibility of the soft tissue structures. The device may include a handle extending through the body.

(25) The subject invention, however, in other embodiments, need not achieve all these objectives and the claims hereof should not be limited to structures or methods capable of achieving these objectives.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

- (1) Other objects, features and advantages will occur to those skilled in the art from the following description of a preferred embodiment and the accompanying drawings, in which;
- (2) FIG. 1 is a three-dimensional perspective side-view of one embodiment of the therapeutic, fitness, and sports enhancement device of this invention;
- (3) FIG. 2 is a schematic end-view showing in further detail one embodiment of the projections shown in FIG. 1;
- (4) FIG. 3 is another three-dimensional perspective side-view of the therapeutic, fitness, and sports enhancement device shown in FIGS. 1 and 2;
- (5) FIG. 4 is a schematic end-view showing another embodiment of the projections of this invention;
- (6) FIG. 5 is a schematic end-view showing an example of the shaped projections shown in FIG. 4 extending at different angles;
- (7) FIG. 6 is a schematic end-view of another embodiment of the projections of this invention;
- (8) FIG. 7 is a schematic end-view of yet another embodiment of the projections of this invention;
- (9) FIGS. 8A-8E are schematic views showing additional embodiments of the projections of this invention;
- (10) FIGS. 9A and 9B are schematic top-views showing exemplary individual patterns of the projections of this invention;
- (11) FIG. 10 is a three-dimensional side view showing three different examples of the therapeutic, fitness, and sports enhancement device of this invention having different densities, diameter, and shaped projections;
- (12) FIGS. 11A-11C and 11E-11M are three-dimensional views showing examples of one embodiment of the therapeutic, fitness, and sports enhancement device shown in FIGS. 1-3 and 10 used to mobilize exemplary soft tissue structures of the human body in accordance with this invention;
- (13) FIGS. 12A-12I are three-dimensional views showing examples of another embodiment of the therapeutic, fitness, and sports enhancement device shown in FIGS. 1-3 and 10 used to mobilize different exemplary soft tissue structures of the human body in accordance with this invention;
- (14) FIG. 13 is a schematic side-view showing one example of an extrusion process which may be used to manufacture the therapeutic, fitness, and sports enhancement device shown in FIGS. 1-10;
- (15) FIG. 14 is a schematic side-view showing one example of a molding process which may be used to manufacture the therapeutic, fitness, and sports enhancement device shown in FIGS. 1-10;
- (16) FIG. 15 is a schematic end-view showing one example of a plurality of bands which may be used to manufacture one embodiment of the therapeutic, fitness, and sports enhancement device of this invention;
- (17) FIG. 16 is a three-dimensional perspective side-view of another embodiment of the therapeutic, fitness, and sports enhancement device of this invention;
- (18) FIG. 17 is a three-dimensional perspective end-view of the therapeutic, fitness, and sports enhancement device shown in FIG. 16;
- (19) FIG. 18 is a three-dimensional perspective side-view showing one example of end caps which may be utilized with the therapeutic, fitness, and sports enhancement device shown in FIGS. 1-16 of this invention;
- (20) FIG. 19A is a schematic side-view showing one embodiment of the therapeutic, fitness, and sports enhancement device of this invention with a handle;
- (21) FIG. 19B is a schematic end-view of the therapeutic, fitness, and sports enhancement device shown in FIG. 18A; and
- (22) FIG. 19C is another schematic end-view of the therapeutic, fitness, and sports enhancement device shown in FIG. 18A.

DETAILED DESCRIPTION OF THE INVENTION

(23) Aside from the preferred embodiment or embodiments disclosed below, this invention is capable of other embodiments and of being practiced or being carried out in various ways. Thus, it is to be understood that the invention is not limited in its application to the details of construction and the arrangements of components set forth in the following description or illustrated in the drawings. If only one embodiment is described herein, the claims hereof are not to be limited to that embodiment. Moreover, the claims hereof are not to be read restrictively unless there is clear and convincing evidence manifesting a certain exclusion, restriction, or disclaimer.

(24) There is shown in FIG. 1, one embodiment of therapeutic, fitness, and sports enhancement device **10** of this invention including cylindrically shaped body **12** having a predetermined density and predetermined diameter. Body **12** includes a plurality of projections **14** of a predetermined shape. One or more of the predetermined density and the predetermined diameter of body **12** in combination with the shape of the projections effectively mobilize soft tissue structures of the human body, as discussed in further detail below.

(25) Body **12** with projections **14** is preferably made of a pliable material, e.g., a closed-cell foam material, e.g., a chemically, cross-linked closed-cell polyethylene foam, such as MINICEL® L200, L300 or L380, available from Sekisui Voltek (Lawrence, MA). In other examples, body **12** may be made of expandable polyethylene (EPE) or a plastic type material using foam fabricators, as known by those skilled in the art. Other similar type of pliable materials and closed-cell foams known to those skilled in the art may also be utilized to manufacture device **10**.

(26) In one example, the density of body **12**, the diameter, d-**13**, FIG. 2, of body **12** in combination with and the shape of projections **14** are preferably chosen to effectively mobilize soft tissue structures of the human body. The soft tissue structures mobilized in one embodiment may include, inter alia, the Quadriceps, Hamstrings, Rhomboids, Mid/Lower Back, Glutes/Piriformis, Hip Rotators, Hamstrings, IT band, Calf, and Shins. In this example, which may be referred to as the “soft” device **10**, the density of the pliable material of body **12** is preferably in the range of about 2.0 lb/ft.^{sup.3} to about 6.0 lb/ft.^{sup.3}. The diameter, d-**27**, is typically in the range between about 3 inches to about 15 inches, e.g., about 7 inches. In this example, cylindrically shaped body **12**, FIG. 3, preferably has a length, **1-31**, between about 18 inches and 36 inches. In one preferred example, body **12** has a diameter of about 7 inches and a length of about 18 inches. In another example, body **12** has a diameter of about 7 inches and a length of about 36 inches.

(27) The shape projections **14** may be rounded, as shown at **17**, FIGS. 2 and 3. In other designs, the shape of projections **14** may have triangle-shaped, as shown in FIG. 4. In this example, the projections **14** may radially extend from body **12** at the same angle. In other examples, projections **14**, FIG. 5, may radially extend from body **12** at different angles, as indicated at **20**, **22**, and **24**. In other examples, projections **14**, FIG. 6, may be cylindrically shaped with rounded ends or may also be cylindrically shaped with flat ends, as shown in FIG. 7. Projections **14**, FIGS. 1-7, may also extend over the length, **1-31**, FIGS. 1 and 3 of body **12**, e.g., in a right angle pattern, as shown in FIG. 8A, in a helical pattern, as shown in FIG. 8B, in a wave pattern, as shown in FIG. 8C, in a Z pattern, as shown in FIG. 8D, or in a pattern perpendicular to longitudinal axis **15**, as shown in FIG. 8E. In other designs, the plurality of shaped projections **14** may extend over body **12** in an individual pattern as shown in FIGS. 9A and 9B. Each of the plurality of shaped projections **14**, FIGS. 1-9B, preferably radially, extends about 1/8 inch to about 3 inches from body **12**, e.g., as shown by r-**70**, FIG. 2. In one example, “soft” device **10**, FIGS. 1-3, may include eleven projections **14**. In another example, “soft” device **10**, Fig. 10, indicated at **41**, may have twelve projections **14**. In other examples, device **10** may have as few as 6 projections or as many as 20 projections **14**, although device **10** may have any number of projections of various shapes, disclosed above, depending on the therapeutic use of device **10**.

(28) In another example, one or more of the density of body **12** and the diameter of body **12** in combination with the shape of projections **14** are preferably chosen to mobilize soft tissue

structures of the human body which may include, inter alia, the Quadriceps, Hamstrings, Rhomboids, Rotator cuff, Glutes/Piriformis, Iliotibial band, Achilles tendon, Calf, feet, Head and Neck. In this example, often referred to as “firm” device **10**, the density of the pliable material of body **12** is in the range of about 2.0 lb/ft.^{sup.2} to about 12 lb/ft.^{sup.3}, e.g., preferably about 4 lb/ft.^{sup.3}. The diameter, d-**27**, FIG. 2, of body **12** is in the range of about 3 inches to about 8 inches, e.g., preferably about 3.5 or 7 inches. The length of body **12**, length **1-31**, FIG. 3, is in the range of about 2 inches to about 36 inches. In one preferred example, body **12** has a diameter of about 3 inches and a length of about 8 inches. In yet another example, body **12** has a diameter of about 3.5 inches and a length of about 12 inches. In yet another example, body **12** has a diameter of about 7 inches and a length of about 18 inches. In another example, body **12** has a diameter of about 7 inches and a length of about 36 inches. Similar as discussed above, the plurality of shaped projections **14** have a predetermined shaped as discussed above with reference to FIGS. **1-10** and preferably radially extend about ⅛ inch to about 3 inches from body **12**. FIG. **10** shows one example of “firm” device **10**, indicated at **43**. In this example, projections **14** may have a triangle shape as shown. In this design, device **10** has about thirteen projections **14**, although device **10** could have any number of projections of various shapes, discussed above, depending on the therapeutic use of device **10**.

(29) In yet another example, the one or more of density of body **12** and the diameter of body **12** in combination with the shape of projections **14** are preferably chosen to mobilization soft tissue structures of the human body which may include, inter alia, the Quadriceps, Hamstrings, Rhomboids, Upper Trapezius, Forearm, Rotator cuff, Glutes, Piriformis, Calf, Feet, Head and Neck. In this example, which may be referred to as “hard” device **10**, the density of the pliable material of body **12** is in the range of about 4.0 lb/ft.^{sup.2} to about 12 lb/ft.^{sup.3}, e.g., preferably about 6 lb/ft.^{sup.3}. The diameter, d-**27**, FIG. 2, of body **12** is in the range of about 3 inches to about 8 inches, e.g., preferably about 3.5 or 7 inches. The length of body **12**, length **1-31**, FIG. 3, is in the range of about 2 inches to about 36 inches. In one preferred example, body **12** has a diameter of about 3 inches and a length of about 8 inches. In yet another example, body **12** has a diameter of about 3.5 inches and a length of about 12 inches. In another example, body **12** has a diameter of about 7 inches and a length of about 18 inches. In another example, body **12** has a diameter of about 7 inches and a length of about 36 inches. Similar as discussed above, the plurality of shaped projections **14** have a predetermined shaped as discussed above with reference to FIGS. **1-10** and preferably radially extend about ⅛ inch to about 3 inches from body **12**. FIG. **10** shows an example of device **10**, indicated at **45**. In this example, device **10** may include about seven projections with a curved shape as shown, although device **10** may have any number of projections of various shapes, discussed above, depending on the therapeutic use of device **10**.

(30) In the examples above, one or more of the predetermined density and the predetermined diameter in combination with the predetermined shape of the projections effectively mobilize soft tissue structures of the human body. In other examples, one or more of the predetermined density and predetermined diameter in combination with the predetermined shape of the projections may be chosen to effectively optimize body core strength and optimize balance training and/or efficiently break up collagenous fibers of soft tissue structures and/or enhance flexibility of soft tissue structures and/or enhance distensibility of soft tissue structures. Moreover, projections **14** stabilize device **10** when used for body core strength and balance training.

(31) Examples of the operation of therapeutic, fitness and sports enhancement device **10** with projections **14** of one or more embodiments of this invention used to mobilize soft tissue structures of the human body and/or effectively optimize body core strength and optimize balance training and/or efficiently break up collagenous fibers of soft tissue structures and/or enhance flexibility of soft tissue structures and/or enhance distensibility of soft tissue structures are shown in FIGS. **11A-11M**. For example, FIG. **11A** shows an example of device **10** being used to mobilize the soft tissue of Quadriceps **98**. When moved back and forth in direction **100**, projections **14** extend into the soft

tissue of Quadriceps **98** and mobilize Quadriceps **98**. This breaks up the collagenous fibers of the Quadriceps **98**, enhances flexibility and enhances distensibility of Quadriceps **98**.

(32) In other examples, therapeutic, fitness, and sports enhancement device **10** may be used in a similar manner as discussed above with reference to FIG. **11A** to mobilize soft tissue structures of the Iliotibial band (ITB), as shown in FIG. **11B**, the inner thigh, as shown in FIG. **11C**, the Glutes, as shown in FIG. **11E**, the Piriformis, as shown in FIG. **11F**, the Hamstrings, as shown in FIG. **10G**, the Lumbar region, as shown in FIG. **11H**, the Rhomboids, as shown in FIG. **11I**, the Thoracic region (Mid/Lower Back), as shown in FIGS. **11J** and **11K**, the Shins, as shown in FIG. **11L**, and the calf, as shown in FIG. **11M**.

(33) Examples of the operation of “firm” and/or “hard” therapeutic, fitness, and sports enhancement device **10** used to mobilize soft tissue structures of the human body, break up collagenous fibers and/or enhance flexibility and/or enhance distensibility are shown in FIGS. **12A-12L**. For example, device **10** may be used to effectively mobilize soft tissue, break up collagenous fibers and/or enhance flexibility and distensibility of the Neck, as shown in FIG. **12A**, the upper Neck, as shown in FIG. **12B**, the upper Trapezius, as shown in FIG. **12C**, the shoulder and Rotator cuff, as shown in FIG. **12D**, the Rhomboids, as shown in FIG. **12E**, the Lower Back, as shown in FIG. **12F**, the arch of the foot, as shown in FIG. **12G**, the Achilles, as shown in FIG. **12H**, and the Calf, as shown in FIG. **12I**.

(34) The result is therapeutic, fitness and sports enhancement device **10** with a unique combination of one or more of density and diameter and the shape of projections **14** maximizes mobilization of soft tissue structures. Typical target soft tissue mobilized by the plurality of shaped projections **14** may include muscles, tendons, ligaments fascia of the human body and the like. Mobilization of soft tissue by projections **14** massages soft tissue and breaks up scar tissue. This increases flexibility of soft tissue, decreases muscle soreness, relieves joint stress, decreases neuromuscular hypertonicity, increases joint range motion and extensibility of musculotendinous junctions, optimizes soft tissue distensibility, increases neuromuscular efficiency, increases flexibility and helps maintain normal functional muscle length. These benefits enhance recovery of injuries when used as a therapeutic tool, maximize sports performance, and improved fitness. Device **10** also optimizes body core strength and optimizes balance training.

(35) In one example, therapeutic fitness and sports enhancement device **10**, FIGS. **1-12I** may be manufactured using extrusion process as know by those skilled in the art. In this example, raw material in the form of granules of plastic, e.g., polyethylene, is put hopper **150**, FIG. **13**, and fed into heated chamber **152**. Press **154** pushes the material in chamber **152** through die **156**. Die plate has the shape of body **12** with projections **14**. In this example, die **14** has a shape similar to body **12**, FIGS. **1-3**, with rounded projections **14**. In other examples, die **156** can be shaped with any configuration to manufacture device **10** with the projections shaped as shown in FIGS. **1-9B**, or any other various shape known by those skilled in the art. The extruded material is then allowed to cure in a controlled temperature room to strengthen the material and provide a more finished look to device **10**.

(36) In another example, therapeutic fitness and sports enhancement device **10**, FIGS. **1-12I** may be manufactured using molding process, as known to those skilled in the art. In this example, raw material, e.g., beads of expandable polyethylene (EPE), is put into a hopper **160**, FIG. **14**, and then fed into chamber **162** which includes a die in the shape of device **10**, FIGS. **1-12I**. The chamber is heated and chamber **162** is pressurized. This allows the beads of EPE to shape to the contours of the die. Then a curing phase occurs to allow the EPE form the shape of device **10**, e.g., as shown in FIGS. **1-9B**. This curing phase occurs in a heated room to allow the polymer to cure.

(37) In another example, therapeutic fitness and sports enhancement device **10**, may be manufactured with a large sheet of the pliable material is cut using a high speed vertical band saw which forms cylindrically shaped body **12** and the plurality of shaped projections **14** thereon.

(38) In yet another example, device **10** may include cylindrically shaped body **12**, FIG. **15** that

includes a plurality of bonded sections, e.g., sections **19**, **21**, **23**, and **25**, each having a plurality of shaped projections **14** thereon of a predetermined type, e.g., as shown in any of FIGS. **1-9B**. Each of sections **19-25**, FIG. **15**, is similarly made of a pliable material, e.g., closed-cell polyethylene foam, such as MINICEL® L200, L300 or L380 discussed above, and is bonded to each other using a high bonding adhesive. The sheets of the pliable material used for the plurality of bonded sections, e.g., sections **19-25**, typically has predetermined width, e.g., about 2 inches to 3 inches, or up to 4 inches, although the sheets of pliable material and the resulting width of sections **19-25** may have any width as known to those skilled in the art. The shape of the plurality of shaped projections **16** and cylindrically shaped body **12** made of sections **19-25** is similarly formed using a high speed versatile band saw, as discussed above.

(39) In another embodiment, therapeutic, fitness, and sports enhancement device **10**, FIG. **16**, of this invention includes cylindrically shaped core **12** and overlay **13** about core **12**. Similar as discussed above, one or more of the predetermined density and the predetermined diameter of body **12** and overlay **13** in combination with the shape of the projections are chosen to effectively mobilize soft tissue structures of the human body. Overlay **13** includes a plurality of shaped projections **14**, similar as discussed above with reference to FIGS. **19B**, which are shaped to mobilize soft tissue structures of the human body and/or improve body core strength and balance training and/or enhance flexibility and/or optimize soft tissue distensibility. Core **12** is similarly made of pliable material, as discussed above. Overlay **13** is typically made of MINICEL® L200, L300 or L380. Body **12** and overlay **13** with projections **14** thereon may also be formed from an extrusion process or a die cast process or cut by a high speed foam band saw, as known to those skilled in the art. Similar as discussed above with reference to FIG. **2**, projections **14** on overlay **13**, FIG. **16**, radially extends about $\frac{1}{8}$ inches to about 3 inches from overlay **13**, e.g., as indicated at r-**70**, FIG. **2**, as needed to maximize soft tissue mobilization, and improve balance and core strength training, enhance flexibility and optimize soft tissue distensibility. Therapeutic, fitness, and sports enhancement device preferably has a density and d**10'** may have a length as indicated by **1-31**, FIG. **16**, and has a diameter, d-**27**, FIG. **17**, similar to any of the embodiment of device **10** discussed above with reference to FIGS. **1-9B**.

(40) In one embodiment, therapeutic, fitness, and sports enhancement device **10**, FIGS. **1-16**, may include end caps **212**, FIG. **18**, on each end of device **10**.

(41) Therapeutic fitness, and sports enhancement device **10**, FIGS. **1-18**, may also include handle **300**, FIG. **19A**. Handle **300** preferably extends through opening **302**, FIG. **19B**, having inner diameter d-**304**, which can vary from about $\frac{1}{4}$ inch to about 5 inches, e.g., about 2 inches as shown in FIG. **19C**. Opening **302**, FIG. **19B**, with inner diameter d-**304** is typically created using rut extrusion process. Inner sleeve **306** fits inside opening **302** and handle **300** extends through inner sleeve **306**. Handle **300** may be used for gripping to allow the ridged roll to roll over the soft tissue of the body, e.g., the Quadriceps, Upper Trapezius, Iliotibial band, and the like, as discussed above with reference to FIGS. **11A-12I**.

(42) Although specific features of the invention are shown in some drawings and not in others, this is for convenience only as each feature may be combined with any or an of the other features in accordance with the invention. The words “including”, “comprising”, “having”, and “with” as used herein are to be interpreted broadly and comprehensively and are not limited to any physical interconnection. Moreover, any embodiments disclosed in the subject application are not to be taken as the only possible embodiments. Other embodiments will occur to those skilled in the art and are within the following claims.

(43) In addition, any amendment presented during the prosecution of the patent application for this patent is not a disclaimer of any claim element presented in the application as filed: those skilled in the art cannot reasonably be expected to draft a claim that would literally encompass all possible equivalents, many equivalents will be unforeseeable at the time of the amendment and are beyond a fair interpretation of what is to be surrendered (if anything), the rationale underlying the

amendment may bear no more than a tangential relation to many equivalents, and/or there are many other reasons the applicant can not be expected to describe certain insubstantial substitutes for any claim element amended.

Claims

1. A hand-held device, comprising: a single body having a longitudinal length and a plurality of individual projections extending along the longitudinal length of the single body, the plurality of individual projections configured to extend into the muscle below skin of a human body, the plurality of individual projections are spaced apart from each other and are arranged along longitudinally extending lines that are circumferentially spaced apart from each other on the single body, the plurality of individual projections radially extend in the range from about $\frac{1}{8}$ inch to about 3 inches, the single body having an opening extending therethrough along the longitudinal length of the single body; and a handle extending through the opening such that a first handle grip is on one side of the single body and a second handle grip is on the opposing side of the single body, the first and second handle grips made for being grasped by a user when moving the hand-held device over the human body such that each of the plurality of individual projections move in unison in the circumferential direction around the handle.
2. The device of claim 1, wherein the single body with the plurality of individual projections is made of at least one of a closed-cell foam, a polymer, or a rubber.
3. The device of claim 2, wherein the density of the single body is between about 2 lbs/ft.³ and about 12 lbs/ft.³.
4. The device of claim 1, further including a sleeve that fits within the opening, the handle extends through the sleeve.
5. The device of claim 1, wherein the plurality of individual projections have a predetermined shape including one of a rounded shape, a curved shape, a triangle shape, a square shape, a notch shape, a cylindrical shape with rounded ends, or a cylindrical shape with flat ends.
6. The device of claim 1 wherein the plurality of individual projections longitudinally extend over the single body to form an offset pattern.
7. The device of claim 1 wherein the plurality of individual projections longitudinally extend over the single body to form an aligned pattern.
8. The device of claim 1 wherein the plurality of individual projections are the same size.
9. The device of claim 1, wherein some of the plurality of individual projections radially extend from the body at different angles than other ones of the plurality of individual projections.
10. The device of claim 1, wherein the plurality of individual projections cover the entire longitudinal length of the single body.
11. A hand-held device, comprising: a handle having a first handle grip and a second handle grip, the first and second handle grips made for being grasped by a user when moving the hand-held device over the human body, the handle extending along a longitudinal direction of the hand-held device; and a plurality of individual projections located between the first handle grip and the second handle grip, the plurality of individual projections radially extend in a range of about $\frac{1}{8}$ inch to about 3 inches, the plurality of individual projections are spaced apart from each other in the longitudinal direction and have rounded end shapes, the plurality of individual projections are configured to extend into the muscle below skin of a human body, and wherein some of the plurality of individual projections radially extend at different angles than other ones of the plurality of individual projections.
12. The device of claim 11, further including a sleeve that is circumferentially surrounded by the plurality of projections, the handle extends through the sleeve.
13. The device of claim 11, wherein the plurality of individual projections are formed from a closed-cell foam, a polymer or a rubber.

14. The device of claim 11, wherein an outer circumference of each of the plurality of individual projections has a circular shape.

15. The device of claim 11, wherein each of the plurality of individual projections move in unison in the circumferential direction around the handle.

16. A hand-held device, comprising: a single body having a longitudinal length and a plurality of individual projections extending along the longitudinal length of the single body, the plurality of individual projections configured to extend into the muscle below skin of a human body, the plurality of individual projections are spaced apart from each other in a longitudinal direction of the single body, each of the plurality of individual projections has an outer circumference that has a circular shape, the single body having an opening extending therethrough along the longitudinal length of the single body; and a handle extending through the opening such that a first handle grip is on one side of the single body and a second handle grip is on the opposing side of the single body, the first and second handle grips made for being grasped by a user when moving the hand-held-soft tissue mobilization device over the human body such that each of the plurality of individual projections move in unison in the circumferential direction around the handle.

17. The device of claim 16, wherein the density of the single body is between about 2 lbs/ft.³ and about 12 lbs/ft.³.

18. The device of claim 16, wherein each of the plurality of individual projections has a rounded end shape that defines the outer circumference.

19. The device of claim 16, further including a sleeve that fits within the opening, the handle extends through the sleeve.
